31 — Upgrade Paths: Sorption Cycles & Heat Pumps on the Platform Chemistry

v1.0 — boost modes riding on the passive baseline; the $CaCl_2$ absorption heat transformer (X12) as the primary path

Function: Record the vetted family of heat-pump and sorption-cycle additions to the platform — what each one is, what it costs in complexity, what it unlocks, and the cheapest experiment that gates it. Shared basis, design point, and doctrines → doc 00; liquid-track hardware → docs 10–12; solid-track hardware → docs 20–22.

Standing rule (binding for this document): every path here is an **addition or upgrade, never core**. Each one introduces something the baseline deliberately excludes — an electrical dependence, a refrigerant circuit, a vacuum vessel, or a crystallization-adjacent process. No DP-A comfort, air-quality, or water claim in docs 00–30 may be made to depend on any of them; they ride *on top of* the passive design as named boost modes, and the baseline must remain fully operable with every one of them removed.

Status of numbers: estimate-grade throughout unless marked, PENDING the named gates. Nothing in this document has been bench-tested.

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1. The family at a glance — and one recorded rejection

#	Path	Character	Converts	Into	First gate
§2	**$CaCl_2$ absorption heat transformer (X12)**	heat-only, no new chemistry	60–65 °C waste tail	85–90 °C process heat	**test L**
§3	Coupled absorber–regen VC heat pump	electron-rich boost	~350–500 W electric	brine floor depth + regen heat	procurement + test I data
§4	MVR on the sealed still	heat-outage regenerator	~110–140 W electric	full still duty, no heat source	compressor mist pre-gate (test-B analogue)
§5	Closed AlFu/water adsorption chiller	heat-rich boost	2–2.5 kW solar/waste heat	heat-driven brine chilling	solid-track M-series + vacuum leak-down
§6	Static crystallizer pot (+ hexahydrate PCM note)	mass-limited platforms	reserve tank mass	3–5× denser moisture battery	**test A3**

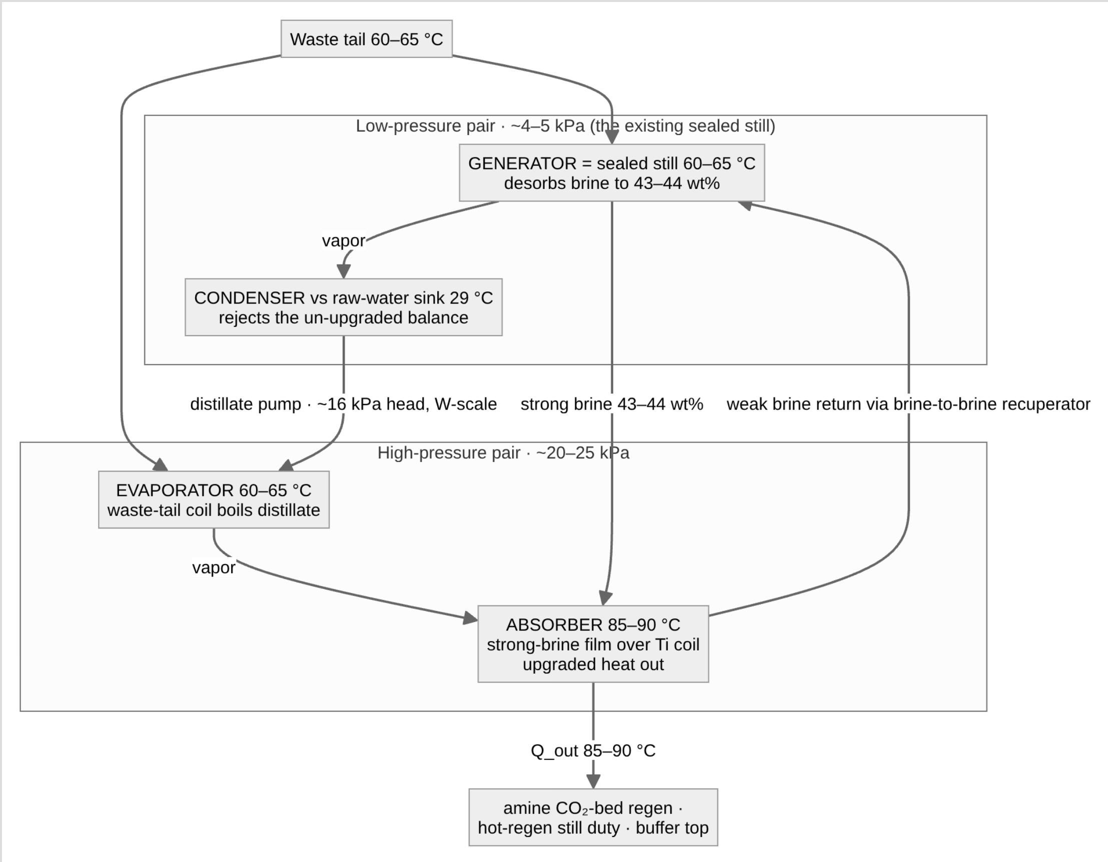
Recorded rejection: a heat pump as *prime mover* — lifting from the 29 °C raw-water sink to regeneration grade as the system's primary heat source — is rejected. At a ~35 K sink-to-regen lift the real COP_h is ~3.5–4, costing ~2 kW electric at solid-track duty; that loses outright to a 1–3 kW vapor-compression AC and abandons the heat-flexibility value proposition (doc 20 §9). Heat pumps earn a place on this platform **only as topping lifts on nearly-hot heat** (§3) — never as the engine.

2. X12 — the $CaCl_2$ absorption heat transformer (primary upgrade)

2.1 The finding

X12: the desiccant's activity depression is a temperature lift, not only a humidity floor. Water vapor absorbed into concentrated brine releases its latent heat (~2.6 MJ/kg) *at the brine's temperature* — and because the brine's vapor pressure is depressed by aw, absorption stays thermodynamically downhill even when the brine sits 20–25 K **hotter** than the vapor's source. The platform therefore already stocks every component of a single-stage **type-II absorption heat transformer (AHT)**: the working pair is the existing $CaCl_2$/water system, and the low-pressure half of the machine — the generator and its condenser — **is the sealed still of doc 11 §4, running unchanged at its design condition**. X12 is the dual of X2: X2 says the still's lack of a purge keeps solar-grade *desorption* alive; X12 says the same activity depression keeps above-source-temperature *absorption* alive.

2.2 The cycle — four vessels, two pressure levels



Walking the loop: (1) the still desorbs at 60–65 °C against its sink-cooled condenser exactly per doc 11 §4, producing 43–44 wt% strong brine and distillate. (2) A peristaltic pump lifts distillate to the high-pressure pair — a ~16 kPa head, so unlike LiBr machines the "solution pump" is a few watts of existing hardware. (3) In the evaporator, more of the same 60–65 °C tail boils that distillate at ~20–25 kPa. (4) The vapor flows to the absorber, where strong brine filmed over a titanium extraction coil absorbs it at 85–90 °C, releasing ~2.6 MJ/kg into the coil — the useful, upgraded output. (5) Diluted brine drains back to the still through a brine-to-brine recuperator and the loop closes. Roughly half the driving heat emerges upgraded; the balance leaves through the still condenser the platform was already running.

2.3 Governing inequality and the lift ceiling

Absorption at the high-pressure level requires

> **aw(x) · P_sat(T_abs) < P_sat(T_evap)**

which caps the absorber temperature for a given concentration and evaporator temperature:

Strong brine	aw	T_evap 60 °C (~20 kPa)	T_evap 65 °C (~25 kPa)
44 wt%	~0.34	T_abs ≤ 85–86 °C	**T_abs ≤ 91 °C**
42 wt%	~0.40	T_abs ≤ 81 °C	T_abs ≤ 86–87 °C

The honest deliverable is a **gross lift of ~20–25 K into the 85–90 °C band** — precisely the amine CO_2-bed regeneration grade (doc 00 §5) and the hot-regen still grade (doc 11 §4), the two loads currently stranded above a 60–65 °C waste tail. The ceiling is real and single-stage: the X11 potash rung (~130–150 °C) is **beyond reach** — a two-stage AHT could lift further in principle but is not pursued (complexity discipline; the potash bed remains ≥130 °C-tap hardware only). The aw table above carries the same ±0.05 uncertainty as everything else on the brine side — **test A's measured aw table feeds this inequality directly.**

2.4 Performance and a sizing example

Single-stage AHT COP (upgraded heat ÷ driving heat) runs **~0.45–0.48** before losses. Concretely: delivering the amine CO_2 battery's ~2–2.5 kWh/day at 85–90 °C draws **~5 kWh/day of 60–65 °C tail** and rejects the balance through the existing still condenser. Continuous form: ~1 kW of upgraded heat per ~2.2 kW of tail. On a waste-heat platform whose tail is free, the COP is economically irrelevant (doc 10 §3's "COP stops mattering" rule) — what the AHT buys is **grade**, with zero electricity beyond the transfer pumps.

What it unlocks, by load:

- **Amine CO_2-bed regeneration (85–95 °C) from a 60–65 °C-only platform** — closing the one gap in the X11 ladder where a waste tail exists but tops out below amine grade, without touching the solar path.
- **Hot-regen brine (43–44 wt% → 7.5–9 g/kg floor)** where the heat bus never exceeds 65 °C — the deep-dry lever without LiCl and without 85–93 °C primary heat.
- **Buffer-top support** (doc 11 §7): trims into the 65–90 °C band from below.

2.5 Hardware mapping — what is new and what is not

AHT component	Platform part	Status

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Generator + condenser	**The doc 11 §4 sealed still, unchanged**	existing design
Solution/transfer pumps	Peristaltic (doc 11 §6)	existing class
Evaporator	New sealed tray/vessel + waste-tail coil, ~20–25 kPa	**new** — still-class construction, higher pressure than the still (easier vacuum)
Absorber	New vessel: strong-brine film distribution over a Ti extraction coil	**new** — the one genuinely novel part
Recuperator	Brine-to-brine, doc 00 §7 rule-4 class	new but conventional

2.6 Materials and the crystallization inversion

The absorber runs 85–90 °C **brine** — past comfortable CPVC territory (93 °C ceiling with no margin). **PVDF or PP-with-margin for the absorber vessel and internals; titanium for the extraction coil**; everything inside the two-worlds law (doc 11 §1), which is untouched. The crystallization risk is inverted from the tropical norm: 44 wt% is safely liquid *hot*, so the hazard is **cooldown in the absorber and its lines after shutdown** (44 wt% liquidus ≈ 22 °C is not the issue at tropical ambient — but concentration excursions and cool-climate land installs are). Design response, extending safety-register item 6: **drain-back-to-still on stop** — the absorber and its lines empty to the low-pressure sump whenever transfer stops, so no static high-concentration inventory can cool in place.

2.7 Gating test L — hot-film absorption (~$60–120)

The literature K·a values for brine films are near-ambient; nothing published speaks to film absorption at 85–90 °C against ~20–25 kPa steam. Test L is the hot-absorption analogue of test A + I: a small 43–44 wt% film over a heated coupon in a sealed vessel fed with ~60–65 °C steam; measure absorption rate and temperature approach vs film temperature (85 / 88 / 90 °C setpoints), and exercise the drain-back-on-stop behavior including a deliberate cool-in-place fault. Reuses the sous-vide circulator and the wet/dry-bulb + K-type stack (doc 22 §7). **Go/no-go: measurable absorption at 88–90 °C with the test-A aw table confirming the §2.3 inequality's margin.** Runs only when a waste-heat platform variant is actually in play — it does not join the baseline parallel set.

3. Coupled absorber–regenerator VC heat pump (electron-rich boost)

Evaporator on the absorber's brine loop, condenser on the regenerator: both ends useful simultaneously. Operating points fall out of the brine loop — evaporator saturated ~10–15 °C (brine held 18–20 °C through an approach), condenser ~65–70 °C (60–65 °C regen water into the buffer). A ~55 K lift with a hot condensing end: small heat-pump-water-heater territory.

Refrigerant	Cond. P @ 70 °C	Verdict
R600a (isobutane)	~1.05 MPa	Best thermodynamic fit at high lift; low volumetric capacity → physically larger compressor, fine at 1 kW; A3, charge ~150–300 g
R290 (propane)	~2.6 MPa	**Procurement winner** — 0.5–1 kW HPWH rotaries are R290 off the shelf; T_crit 96.7 °C clears 70 °C condensing
R1234ze(E)	~1.4 MPa	A2L option if flammability governs; small compressors less common
R134a	~2.1 MPa	Works; efficiency droops near 70 °C condensing; relevant because 12/48 V BLDC marine/RV rotaries are R134a

Sizing: evaporator duty = 0.66 kW absorption heat + parasitic ingress from 32 °C air warming chilled brine through the contactor ≈ **0.8–1.0 kW steady**; condenser output ≈ 1.1–1.5 kW, covering the 0.92 kW peak regen duty with surplus into the buffer (DHW-capped per doc 00 §7 rule 4). Real COP_h at this lift with a small hermetic: 2.5–3.0 → **~350–500 W electric**; inverter-driven to track duty. Two-worlds handling: keep the refrigerant circuit entirely standard (copper) and interpose a small glycol loop between evaporator and the Ti sump coil — costs a 2–3 K approach, returning ~1.5–2 g/kg of the gained floor; **net gain at 20 °C brine ≈ +3.5–4.5 g/kg**, i.e., most of the LiCl/hot-regen benefit for ~400 W and no new chemistry. A direct Ti-tube evaporator recovers the approach but is a custom part.

Role: **boost mode** — a 5× step over the 25–80 W baseline, admitted only as a mode the baseline survives without. Cooling leverage per doc 12 §1 (0.5–0.7 g/kg·°C) is the conversion rate; test I's measured K·a at chilled film temperatures refines the gain. Dual-duty note: revalved (condenser coil into the still pool, evaporator on the still's condensate condenser), the same machine is a refrigerant-mediated vapor recompressor — one compressor serving §3 and §4's function with standard hardware.

4. MVR on the sealed still (heat-outage regenerator, ~110–140 W)

Pull vapor off the still headspace, compress, condense in a coil submerged in the pool — the vapor's own latent heat drives further evaporation and external heat input drops to losses. **The lift decomposes into two parts, and the second is the one the literature misses:** at a 70 °C pool over 42 wt% brine (aw ≈ 0.40) the headspace sits at ~12.5 kPa; condensing in-pool with a useful 6–8 K ΔT needs ~40 kPa (sat. 76–78 °C). Pressure ratio ~3 — of which nearly half exists purely to overcome the activity depression. *The machine pays for the desiccant's aw as pressure ratio* (the compression-side mirror of X12). Real specific work ~120–160 Wh/kg vs ~1,800 Wh/kg thermal; at 0.88 kg/h peak duty, **~110–140 W electric** — PV-scale, which is the point: it makes the moisture battery rechargeable on electricity alone through a no-sun heat outage.

Engineering record (honest): sub-atmospheric throughout — non-condensables accumulate in the pool coil and need periodic purging (small vacuum pump or scheduled vent-and-repump). Vapor volume ~11–13 m³/h at 12.5 kPa: small, but PR ~3 excludes regenerative blowers (PR ~1.2 ceiling) and most diaphragm pumps — the realistic machine is a small oil-free scroll or claw pump run as a compressor, or two blower stages. Compression superheats ~80–90 K → a condensate-spray desuperheater ahead of the pool coil protects seals and improves the condensing coefficient. **Pre-gate (test-B analogue): a PP demister ahead of the compressor is mandatory and a downstream coupon must stay clean** — entrained chloride mist is the materials gate; past the demister the stream is essentially clean steam and materials relax to ordinary. Ladder slot: doc 11 §4, below the sealed still, above air-swept — the electricity-rich degraded mode.

5. Closed AlFu/water adsorption chiller (heat-rich brine chilling)

The solid track's chemistry run backward for the liquid track's benefit: instead of drying air, a closed two-shell AlFu/water cycle makes **heat-driven**

chilled water for the brine sump — §3's function with heat instead of electricity. Two sealed shells each hold a DCHX module straight off the doc 22 BOM (same sorbent, coating SOP, and M1–M4 characterization), linked by check-flap vapor paths to a shared raw-water condenser and an evaporator at ~15–18 °C chilling a glycol loop into the Ti sump coil. Beds alternate on ~10-min half-cycles with the doc 22 rig's hot/cold fluid switching, verbatim. Closed cycle → **no purge stream → X2-clean**: it regenerates at genuine solar grade (60–90 °C) — the solid chemistry finally working on solar heat, sidestepping F2 the same way the still does.

Sizing: holding brine at ~20 °C rejects ~0.9–1.0 kW of cooling duty; at a realistic thermal COP of 0.4–0.5, that draws **~2–2.5 kW of 60–90 °C heat** — roughly tripling the liquid track's heat demand while running (+2–3 m² of collector if solar-fed). Strictly a heat-rich boost (engine jacket underway, midday solar surplus). The honest hurdle: the evaporator runs ~1.7–2 kPa absolute — deeper vacuum than §4 — with leak-tightness as the make-or-break property, plus charge degassing and non-condensable purging. Materials science is free (the solid track's); the vessel engineering is not. Gated on the solid M-series regardless; whichever of §3/§5 gets built first is decided by whether the platform is heat-rich or electron-rich.

6. Controlled crystallization — the static crystallizer pot

6.1 The pot (moderate complexity, mass-limited platforms)

$CaCl_2$'s hazards in circulation — the hydrate ladder (6→4→2→1→0, two tetrahydrate polymorphs), 10–20 K supercooling without seeding, caking, slurry in tubing — are exactly what the crystallization interlock exists to prevent. Nearly all of that complexity evaporates under one rule: **crystals never travel.** The tractable form is a dedicated HDPE vessel, thermally and hydraulically isolated from the circulating loop by a normally-closed valve pair, where hot 43–44 wt% concentrate is parked and allowed to deposit dihydrate on cooling. Discharge is dissolution in place — meter in dilute brine or water, stir, draw off liquid at 40 wt%. The circulating loop never sees anything above 43 wt%; the interlock philosophy survives intact because the pot is *designed* to crystallize and everything else remains designed not to.

Payoff: dissolving $CaCl_2 \cdot 2H_2O$ (75.5 wt% $CaCl_2$) to 35 wt% takes up ~1.1–1.2 kg of water per kg of dihydrate vs 0.143 kg/kg for the 40→35 liquid swing — **~3–5× reserve-mass reduction** after mother liquor, shrinking the ~35–40 kg occupied-night charge toward **10–15 kg**. Meaningful where the mass budget binds (marine); convenience elsewhere.

Gating test A3 (~$10, extends test A): seeded vs unseeded jars of 44 wt% parked at ~25 °C — supercooling degree, phase actually formed, caking over ~50–100 park/dissolve cycles, redissolution rate vs stirring. Shares test A's jars and RH/temperature instrumentation.

6.2 Hexahydrate PCM (logged, low priority)

$CaCl_2 \cdot 6H_2O$ melts ~29–30 °C — one of the oldest PCM systems, with its two vices (supercooling, phase segregation) and standard fixes (1–2 wt% $SrCl_2 \cdot 6H_2O$ nucleator, silica/CMC thickener) well documented; static sealed containers, no pumping — genuinely low complexity. The problem is value, not complexity: at DP-A the melt point sits essentially *at* the sink temperature, leaving almost no ΔT to charge or discharge against. It earns its mass only in shoulder climates or paired with a §3/§5 chiller that can freeze it — an accessory to a boost mode, not a standalone item. Logged; no test assigned.

7. Decision record — which first

Platform character	First upgrade	Because
Waste-heat tail ≤65 °C, sealed operation wanted	**§2 AHT (X12)**	Unlocks amine-bed grade and hot-regen floors from heat already free; no new chemistry; the still is half the machine
Electron-rich (big PV/battery), heat-poor	§3 VC heat pump → §4 MVR	~400 W buys LiCl-class floors; ~120 W keeps the moisture battery rechargeable through heat outage; one compressor can serve both
Heat-rich solar, electricity-poor	§5 adsorption chiller	Converts surplus heat to floor depth; waits on the solid M-series anyway
Marine mass-limited	§6 crystallizer pot	3–5× denser moisture battery for one vessel and a $10 jar test

Every path preserves doc 30 §7 principle 1 — redundancy outranks efficiency — because each is severable: the passive baseline beneath it is untouched.

Version history

- **v1.0** — Initial upgrade-paths document: X12 AHT recorded as primary with cycle, lift-ceiling inequality, hardware mapping, drain-back rule, and test L; coupled VC heat pump, still MVR, closed AlFu chiller, and controlled-crystallization paths recorded with gates; prime-mover heat pump rejection recorded; tests L and A3 registered in doc 12 §4.